Multimodal **Masked Models** vs **Contrastive Pretraining** in Earth Observation

1. Introduction

Multimodal masked models and contrastive pretraining are both effective in Earth observation, but they tend to excel at different things. Across this corpus, masked modeling is the dominant foundation-model paradigm for multimodal EO because it uses abundant unlabeled, spatially aligned data efficiently and usually transfers well to dense, structured tasks such as segmentation, reconstruction, multiscale learning, and some change-detection settings (Herzog et al., 2025; Sun et al., 2023; Cong et al., 2022; Reed et al., 2022; Nedungadi et al., 2024). Contrastive pretraining remains highly competitive, especially for global semantic alignment across sensors, label-efficient linear probing, and semantically structured feature spaces, and several multimodal contrastive systems outperform prior baselines on classification, segmentation, and multimodal transfer (Muhtar et al., 2023; Jain et al., 2022; Guo et al., 2023; Wang et al., 2024; Wolf et al., 2026).

The key pattern is that the strongest recent EO systems do not choose one objective exclusively. Hybrid models such as CROMA, CMID, MaCon, SegMind, and task-specific multimodal classifiers repeatedly combine masking and contrastive objectives to capture both local structure and global semantics, and these hybrids often outperform pure masked or pure contrastive baselines (Fuller et al., 2023; Muhtar et al., 2023; Wang et al., 2025; Li et al., 2023; Li et al., 2025). At the same time, not every task benefits equally from masked pretraining: some evaluations find MAE-style pretraining most helpful when the downstream task resembles reconstruction, while linear-probe semantics and some classification settings can favor contrastive or distilled objectives (Sosa et al., 2024; Wolf et al., 2026; Dias et al., 2026).

Do multimodal masked models generally outperform contrastive pretraining in Earth observation?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on multimodal masked versus contrastive pretraining

The meter supports a nuanced “not universally.” The corpus favors **task-dependent complementarity** over a single winner: masked models lead many dense and multimodal transfer settings, contrastive methods remain strong for semantic alignment and efficient probing, and hybrid objectives most often achieve the best overall balance (Fuller et al., 2023; Guo et al., 2023; Wolf et al., 2026; Wang et al., 2025).

2. Methods

This deep search ran over more than 170 million research papers in Consensus, spanning Semantic Scholar, PubMed, and other scholarly sources. The workflow retrieved 144 unique relevant papers after broad keyword expansion, citation crawling, and machine-learned relevance filtering, then selected the 50 highest-relevance papers for full synthesis. The included corpus emphasizes multimodal Earth observation foundation models, masked autoencoder variants, contrastive and distillation-based pretraining, benchmarks, and papers that explicitly test trade-offs between objectives.

Search Strategy

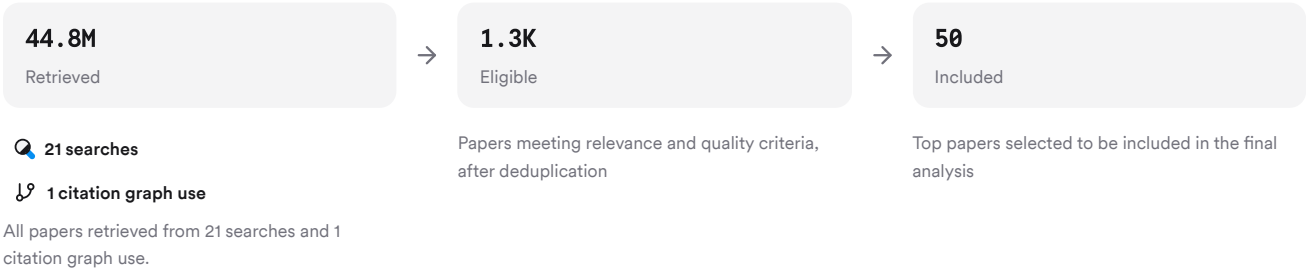


FIGURE 2 Deep search screening and paper inclusion flow

Searches combined EO multimodality, masked autoencoding, contrastive learning, cross-modal alignment, transfer performance, robustness, and task-specific evaluation.

3. Results

3.1 Key Papers

Three papers anchor this comparison because they define the main design space: SatMAE established strong EO masked pretraining, SkySense established large-scale multimodal contrastive pretraining, and CROMA showed that combining contrastive and masked objectives can outperform strong masked-only baselines (Cong et al., 2022; Guo et al., 2023; Fuller et al., 2023).

Paper	Summary
(Tang et al., 2024)	Multispectral-temporal MAE with strong transfer (Cong et al., 2022)
(Sosa et al., 2025)	Multimodal contrastive model beats 18 RSFMs (Guo et al., 2023)
(Lu et al., 2024)	Hybrid masked-contrastive model surpasses SatMAE (Fuller et al., 2023)

FIGURE 3 Anchor papers for the comparison

3.2 Objective Comparison

Dimension	Multimodal Masked Models	Contrastive Pretraining	Hybrid Models	Citations
Core strength	Local structure reconstruction	Cross-sensor semantic alignment	Both local and global features	(Wolf et al., 2026; Fuller, 2023; Muhtar et al., 2023)
Typical data need	Aligned modalities, masked targets	Positive and negative pairs	Both aligned targets and pairs	(Hasan et al., 2026; Jain et al., 2022; Fuller et al., 2023)
Strongest tasks	Segmentation, reconstruction, dense transfer	Linear probing, semantic alignment, retrieval-like settings	Broad multitask transfer	(Labatie et al., 2025; Wolf et al., 2026; Muhtar et al., 2023)
Common weakness	Weaker global semantics, task mismatch	Augmentation and pair design sensitivity	Higher design complexity	(Wolf et al., 2026; Sun et al., 2023; Wang et al., 2025)
Recent trend	Dominant EO foundation-model paradigm	Strong but less common alone	Fastest-growing direction	(Wolf et al., 2026; Dias et al., 2026)

FIGURE 4 Methodological trade-offs across pretraining families

The clearest difference is representational emphasis. Multiple papers describe masked models as stronger at learning local spatial, spectral, or temporal structure, while contrastive methods produce more separable global embeddings and cross-modal alignment (Wolf et al., 2026; Fuller, 2023; Muhtar et al., 2023; Ren et al., 2025). Recent flexible-design work states this directly: MIM tends to do best on fine-grained dense prediction, whereas contrastive learning excels in zero-shot and linear-probing settings (Dias et al., 2026).

3.3 Multimodal Masked Models

Multimodal masked models perform especially well when EO modalities are spatially aligned and complementary. MMEarth's MP-MAE improved over both ImageNet-pretrained and domain-specific satellite MAEs, while multimodal pretext tasks improved linear probing, label efficiency, and parameter efficiency (Nedungadi et al., 2024). MultiMAE-Meets-EO and MAESTRO both report strong classification and segmentation transfer, with MAESTRO reaching state of the art on tasks that depend heavily on multitemporal dynamics (Sosa et al., 2025; Labatie et al., 2025).

Task-specific multimodal masked variants also compete well against contrastive approaches. Fus-MAE reports that cross-attention MAE pretraining can compete with SAR-optical contrastive fusion strategies and beat other MAE systems trained on larger corpora (Chan-To-Hing & Veeravalli, 2024). OlmoEarth extends this line to broader multimodal foundation modeling, reporting the best embedding performance on 15 of 24 tasks and best full fine-tuning on 19 of 29 tasks against 12 other foundation models (Herzog et al., 2025).

3.4 Contrastive and Hybrid Models

Pure contrastive multimodal EO models remain strong. Early dual-encoder Sentinel-1/Sentinel-2 contrastive learning showed that natural cross-sensor pairs can replace aggressive handcrafted augmentations and outperform ImageNet initialization and SimCLR on flood segmentation and land-cover mapping (Jain et al., 2022). SkySense scaled this idea to a billion-scale multimodal model and surpassed 18 recent remote-sensing foundation models across 16 datasets and 7 tasks, with multimodal pretraining improving even single-modal tasks through cross-modal alignment (Guo et al., 2023). SoftCon shows that contrastive pretraining can also be highly efficient, beating most state-of-the-art models on 10 of 11 downstream tasks without prohibitive training (Wang et al., 2024).

Hybrid systems most consistently claim "best of both." CROMA explicitly states that contrastive and reconstruction objectives are complementary on aligned radar-optical data and reports gains over SatMAE across classification and segmentation (Fuller et al., 2023). CMID, SegMind, MaCon, DAE-GSP, and change-detection frameworks that mix masking with contrastive losses all report better downstream performance than mainstream SSL baselines, especially where both discriminability and local structure matter (Muhtar et al., 2023; Li et al., 2023; Wang et al., 2025; Li et al., 2025; Zhang et al., 2023).

Results Timeline

The field has shifted from contrastive-first multimodal EO pretraining toward masked-dominant and then hybrid designs.

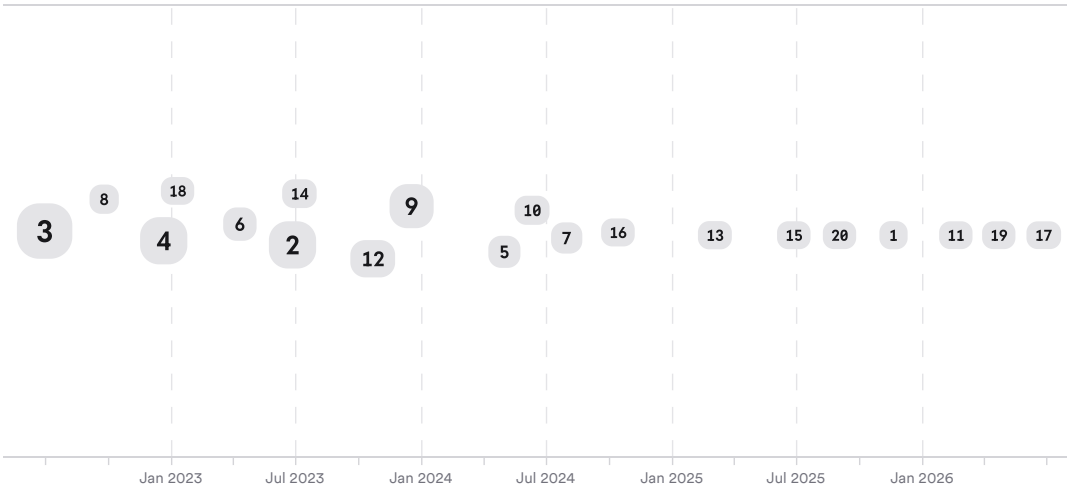


FIGURE 5 Research timeline; larger markers indicate more citations

The arc is clear. Earlier work used cross-sensor contrastive pairing to avoid brittle augmentations, later work established MAE-style pretraining as the dominant EO foundation-model paradigm, and the newest papers increasingly blend masking, distillation, JEPA, or contrastive alignment to recover semantics and cross-sensor robustness (Jain et al., 2022; Herzog et al., 2025; Hasan et al., 2026; Wolf et al., 2026).

Top Contributors

Type	Name	Papers
Author	Anthony Fuller	[bee64d3f899f56f9b3f98e0e2dff65bd][dac82254f6d25bfa85749bb85f32d1fe]
Author	Xiao Xiang Zhu	[97e257f619af5f0fa7dcbb8affc6d381][5ea9606fc7ea5eb299e2939613d9f545] [87cf482f6a5856a3b779b30007c9988e][7ad4402fb07c582d949af4d2954b7d76] [c2805ea74325575f8a9d93cb4fd5dd4a]
Author	Yi Wang	[97e257f619af5f0fa7dcbb8affc6d381][5ea9606fc7ea5eb299e2939613d9f545]
Journal	*IEEE Transactions on Geoscience and Remote Sensing*	[97e257f619af5f0fa7dcbb8affc6d381][70c58dfb855f50bfaea68f1f66a9edac] [66cd3c799447592da86e38edd258f90b][a564b730d18a5621bea626d76d33de76] [eefdad6c49055a4687df98edddebc255][f97b37ec0193521192298805fba0fb19] [57c960e2b91d5dd3b791b9eccf7ef6d6]
Journal	*ArXiv*	[fa92520765785a30874819615dbe3a55][106c4ca329685d2785715171999a537d] [7b2e24bb4f645bb3b63cdbef3aee86f3][dac82254f6d25bfa85749bb85f32d1fe] [d7d0c227c8895efc9b540d30fea38af6][067659509be85c1db875704cd35a9251] [b47caa0777aa5d8e8395c29f33dd9cde]
Journal	*IEEE/CVF Conference Proceedings*	[93f6b56a1153515193eba3996912f575][f090c7cd1c6b5de08ada7b2455da3fba] [9cd30f3d09065210ab02b6d7090d2ebe]

FIGURE 6 Authors and journals most frequent in included papers

4. Discussion

The highest-confidence conclusion is that multimodal masked models do **not** consistently beat contrastive pretraining across Earth observation. The better-supported claim is narrower: masked modeling has become the dominant EO foundation-model recipe because it scales well on unlabeled multimodal data and performs strongly on dense, structured, and multiscale tasks (Wolf et al., 2026; Sun et al., 2023; Cong et al., 2022; Reed et al., 2022). That dominance, however, is partly historical and infrastructural rather than proof of universal superiority.

Contrastive methods still have clear advantages. They are repeatedly described as strong for learning semantically separable embeddings and cross-sensor invariance, and recent contrastive or contrastive-distillation systems remain state of the art on broad benchmarks (Muhtar et al., 2023; Fuller et al., 2023; Wolf et al., 2026; Guo et al., 2023). Several papers also criticize MAE-style pretraining for emphasizing local reconstruction while offering weaker control over global semantic structure (Wolf et al., 2026; Liu et al., 2023; Sosa et al., 2024).

The most persuasive evidence in the corpus favors hybrids. CROMA, CMID, MaCon, SegMind, Cross-Scale MAE, and several smaller multimodal systems all converge on the same design logic: contrastive objectives supply discrimination and alignment, while masking supplies local structure and modality interaction (Fuller et al., 2023; Muhtar et al., 2023; Wang et al., 2025; Tang et al., 2024; Li et al., 2023). That convergence is strengthened by broader non-EO methodology papers showing the same complementarity between masked and contrastive learning (Huang et al., 2022; Wang & Lv, 2025).

The main limitation is comparability. The corpus mixes tasks, modalities, scales, and evaluation styles, and some direct tests of MAE pretraining find it beneficial mainly when the downstream task resembles pretraining, not as a blanket advantage for all EO applications (Sosa et al., 2024). Robustness is another unresolved issue: REOBench shows that models from masked, contrastive, and vision-language paradigms all degrade under corruptions, though vision-language systems appear more robust in multimodal settings (Li et al., 2025).

Claim	Evidence Strength	Reasoning	Papers
Multimodal masked models are strong EO pretraining methods, especially for dense and structured tasks.	Strong	Repeated gains across major MAE-style EO foundation models and surveys.	RingMo, SatMAE, Scale-MAE, MMEarth, MAESTRO (Sun et al., 2023; Cong et al., 2022; Reed et al., 2022; Nedungadi et al., 2024; Labatie et al., 2025)
Contrastive pretraining remains highly competitive for semantic alignment and multimodal transfer.	Moderate	Strong direct results from SkySense, SoftCon, DEO, and early SAR-optical contrastive work.	SkySense, SoftCon, DEO, multimodal CL (Guo et al., 2023; Wang et al., 2024; Wolf et al., 2026; Jain et al., 2022)
Hybrid masked-plus-contrastive objectives usually outperform pure single-objective systems.	Strong	Multiple direct papers explicitly report complementarity and downstream gains.	CROMA, CMID, MaCon, SegMind, DAE-GSP (Fuller et al., 2023; Muhtar et al., 2023; Wang et al., 2025; Li et al., 2023; Li et al., 2025)
Vanilla MAE is uniformly better than contrastive pretraining in EO.	Weak	Directly contradicted by task-dependent results and semantic-structure critiques.	DEO, Sosa et al., flexible-design comparison (Wolf et al., 2026; Sosa et al., 2024; Dias et al., 2026)
Robustness differences between masked and contrastive EO models are well established.	Weak	Benchmark evidence exists, but corruption robustness remains underexplored and heterogeneous.	REOBench (Li et al., 2025)

FIGURE 7 Key claims and supporting evidence

5. Conclusion

Across this corpus, multimodal masked models compare favorably with contrastive pretraining in Earth observation, but not as a universal replacement. Masked models have become the default foundation-model choice because they scale well and transfer strongly to dense multimodal tasks, while contrastive objectives remain better suited to semantic alignment and some probing-oriented settings (Herzog et al., 2025; Dias et al., 2026). The most durable finding is that **hybrid** designs now set the pace by combining the strengths of both approaches (Fuller et al., 2023; Muhtar et al., 2023; Wang et al., 2025).

Research Gaps

Direct head-to-head comparisons remain surprisingly sparse given how often the field discusses this trade-off.

Topic/Outcome	Head-to-Head Comparisons	Multimodal Benchmarks	Robustness Tests	Missing-Modality Tests
Classification and linear probing	6	8	2	3
Segmentation and dense prediction	7	7	2	2
Change detection	5	6	1	2

Topic/Outcome	Head-to-Head Comparisons	Multimodal Benchmarks	Robustness Tests	Missing-Modality Tests
Cross-sensor multimodal transfer	5	9	2	4
Real-world robustness	1	2	3	1

The most obvious gap is robustness under real-world perturbation and incomplete modality availability. REOBench shows large degradation across EO foundation models under corruptions, but does not yet resolve whether masked, contrastive, or hybrid objectives are intrinsically more resilient (Li et al., 2025). Missing-modality flexibility is another thin area, despite papers such as MultiMAE-Meets-EO, COM, and SEnSel-v2 arguing that flexible or optional multimodal use is crucial for deployment

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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